# Windows 11 H08 installation procedure

1 Creating a Linux environment on Windows

1.1 Installing the WSL2

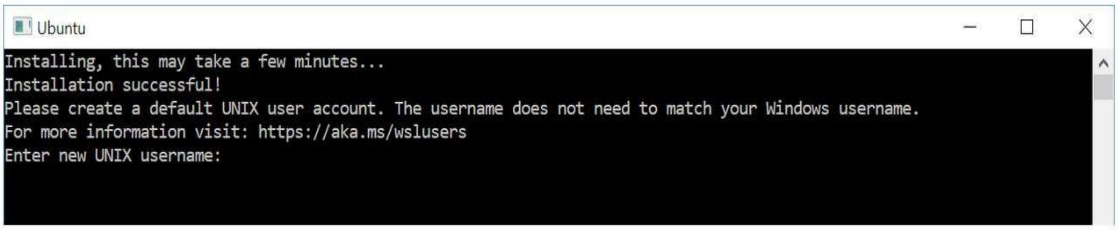
Start Windows PowerShell for administrators from the Start menu and execute the following command.

> wsl --install

Running this command will simultaneously install the latest distribution of Ubuntu on WSL2.

1.2 Creating a Login Account on Ubuntu

When you reboot your PC after installation, Ubuntu will automatically start and ask you to enter the name of the new user account to be created and its password.




1.3 X server (VcXsrv) installed on Windows

(1) Download the latest installer from the following location.

VcXsrv Windows X Server download | SourceForge.net

[https://sourceforge.net/projects/vcxsrv/](https://sourceforge.net/projects/vcxsrv/)

(2) After the download is complete, double-click the file to start the installation. (All selections are OK by default)

(3) Start the X server

A) Double-click the "XLaunch" icon created on the desktop

B) Select "multiple windows" and click "Next.

C) Select "Start no client" and click "Next".

D) Check "Clipboard", "Native opengl", and "Disable access control" and click "Next".

E) When "Finish configuration" is displayed, click "Finish.

F) When the "Windows Security Important Warning" appears, click [Allow Access].

1.4 Ubuntu environment settings (when $SHELL is /bin/bash)

(1) Run ipconfig command in Windows PowerShell

> ipconfig

(2) Get IP of IPv4 address of displayed WSL Ethernet adapter information

(3) Start the Ubuntu terminal and execute the following command

% echo 'export DISPLAY=[IP obtained in (2) above]:0.0' >> ~/.bashrc

% echo 'export PATH=. :$PATH' >> ~/.bashrc

% source ~/.bashrc

2 Pre-installation of relevant software

2.1 Installing gcc, gfortran and cmake

% sudo apt update

% sudo apt upgrade

% sudo -E apt install gcc gfortran cmake

2.2 Installing zlib and szip

% sudo -E apt install zlib1g zlib1g-dev libsz2

2.3 Installing curl and libcurl

% sudo -E apt install curl libcurl4-openssl-dev

2.4 Installing HDF5

% sudo -E apt install hdf5-tools hdf5-helpers libhdf5-dev libhdf5-doc libhdf5-serial-dev

2.5 Installing NetCDF-C and NetCDF-Fortran

% sudo -E apt install netcdf-bin libnetcdf-dev libnetcdff-dev

2.6 GMT Installation

(1) Installation of the related recommended packages

% sudo -E apt install -y ghostscript ffmpeg imagemagick

Then, open the policy file of ImageMagick (/etc/Imagemagick/policy.xml) and disable the ghostscript format types (PS, PS2, PS3, EPS, PDF, XPS).

Example:

<!-- <policy domain="coder" rights="none" pattern="PS" />

<policy domain="coder" rights="none" pattern="PS2" />

<policy domain="coder" rights="none" pattern="PS3" />

<policy domain="coder" rights="none" pattern="EPS" />

<policy domain="coder" rights="none" pattern="PDF" />

<policy domain="coder" rights="none" pattern="XPS" /> -->

※Here, the script is commented out from <!—to -->.

(2) Installing GMT

% sudo -E apt install -y gmt gmt-gshhg gmt-dcw

3 Installing H08

H08 source code is obtained from Github

3.1 Creating your Github account

3.2 Requesting Access to the Global_2018 Repository

Global_2018 is a repository for the distribution of H08 source code.

Please send an email to Tarumoto with the name of your Git account.

You will receive an email invitation to Global_2018, please follow the instructions to accept the invitation.

3.3 Creating a Git Environment on Your Local PC

3.3.1 Installing git

% sudo apt update

% sudo apt upgrade

% sudo apt install git

3.3.2 Setting Up Your Personal Git Environment

(1) User name Setup

% git config --global user.name 'Your Name'

(2) Setting up email

% git config --global user.email 'Your Email Address'

(3) Default Branch Name Setting

% git config --global init.defaultbranch 'main'

(4) Check the settings.

% git config --list

3.3.3 Local configuration for SSH access to Github

(1) Create an access authentication key pair

% ssh-keygen -t rsa -b 4096 -C "[email address]" -f ~/.ssh/git_rsa

(2) Add the following statement to the ~/.ssh/config file

```
--------------------------------------
Host github github.com
      Hostname github.com
      User git
      IdentityFile ~/.ssh/git_rsa
---------------------------------------
```

* Note 1) Copy and paste with indentation.

* Note 2) If necessary, perform the following.

% chmod -R go-rwx ~/.ssh

3.3.4 Github authentication settings for SSH access from local PC

(1) Personal icon in the upper right corner of GitHub Page > Settings > SSH and GPG keys

(2) Click the "New SSH key" button.

(3) Enter a (suitable) key name in the Title field of the displayed page, and copy and paste the contents of the public key (~/.ssh/git_rsa.pub) generated in 3.3.3, (1) into the Key field.

(4) Click the "Add SSH Key" button.

3.4 Copying the Global_2018 Repository to your PC

% cd ~

% git clone git@github.com:organization-h08/Global_2018.git

3.5 Compiling & Installing the H08 Source Code

(1) Change LIB,INC and DC preferences in adm/Mkinclude as follows

INC = -I/usr/include

LIB = -L/usr/lib/x86_64-linux-gnu -lnetcdf -lnetcdff

FC = gfortran -finit-local-zero -O3 -fd-lines-as-comments

(2) Modify the cpl/bin/main.f file as follows

Move the code on line 549 in front of the code on line 336.

Line 549 : character*128 c0lndara

Line 336 : namelist /setriv/ c0qtot, c0rivstoini,

(3) Keep the following line alive in the "# GMT command preference" block of bin/htdraw.sh

```
PSBASEMAP="gmt psbasemap" # ubuntu (if command "gmt" works in your system)
XYZ2GRD="gmt xyz2grd" # ubuntu
GRDIMAGE="gmt grdimage" # ubuntu
PSSCALE="gmt psscale" # ubuntu
PSTEXT="gmt pstext" # ubuntu
PSCOAST="gmt pscoast" # ubuntu
```

(4) Keep the following line alive in the "# GMT command preference" block of bin/htdrawts.sh

```
PSBASEMAP="gmt psbasemap" # ubuntu (if command "gmt" works in your system)
PSTEXT="gmt pstext" # ubuntu
PSXY="gmt psxy" # ubuntu
```

(5) Run the H08 installation script

```
% cd ~/Global_2018
% sh . /install.sh
```

(6) Setting Environment Variables for H08 Execution

```
% cd adm
```

Create a script to modify the sample.bashrc file in this directory to suit your PC environment.

[[ ch_sample.sh ]]

```
------------------------------------------------------------------------------------------
#! /bin/bash

cat sample.bashrc | \
  sed -e 's/PATH=.:${DIRH08}/PATH=${DIRH08}/' | \
  sed -e "s%export PATH=\$PATH:\/sw\/bin%export PATH=${NETCDF}\/bin:\$PATH%" | \
  sed -e "s%export DIRH08=\/Users\/Naota\/H08%export DIRH08=${HOME}\/Global_2018%" \
  > add.bashrc

echo "\n" >> add.bashrc
------------------------------------------------------------------------------------------
```

Run this script.

```
% sh . /ch_sample.sh
```

Add the created add.sample file to ~/.bashrc.

```
% cat add.bashrc >> ~/.bashrc
```

Reflecting the changes.

```
% source ~/.bashrc
```